Configure Interfaces

10.2.1

Configure Router Interfaces

At this point, your routers have their basic configurations. The next step is to configure their interfaces. This is because routers are not reachable by end devices until the interfaces are configured. There are many different types of interfaces available on Cisco routers. For example, the Cisco ISR 4321 router is equipped with two Gigabit Ethernet interfaces:

- **GigabitEthernet 0/0/0 (G0/0/0)**
- **GigabitEthernet 0/0/1 (G0/0/1)**

The task to configure a router interface is very similar to a management SVI on a switch. Specifically, it includes issuing the following commands:

```
Router(config)# interface type-and-number
Router(config-if)# description description-text
Router(config-if)# ip address ipv4-address subnet-mask
Router(config-if)# ipv6 address ipv6-address/prefix-length
Router(config-if)# no shutdown
```

Note: When a router interface is enabled, information messages should be displayed confirming the enabled link.

Although the **description** command is not required to enable an interface, it is good practice to use it. It can be helpful in troubleshooting on production networks by providing information about the type of network connected. For example, if the interface connects to an ISP or service carrier, the **description** command would be helpful to enter the third-party connection and contact information.

Note: The *description-text* is limited to 240 characters.

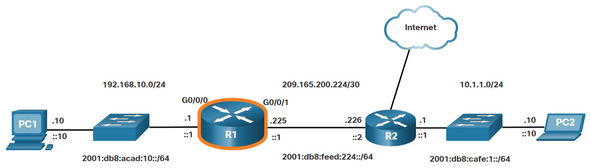
Using the **no shutdown** command activates the interface and is similar to powering on the interface. The interface must also be connected to another device, such as a switch or a router, for the physical layer to be active.

Note: On inter-router connections where there is no Ethernet switch, both interconnecting interfaces must be configured and enabled.

10.2.2

Configure Router Interfaces Example

In this example, the directly connected interfaces of R1 in the topology diagram will be enabled.



To configure the the interfaces on R1, use the following commands.

```
R1> enable
R1# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug  1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Aug  1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Aug  1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
R1(config)#
R1(config)#
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug  1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Aug  1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Aug  1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
R1(config)#
```

Note: Notice the informational messages informing us that G0/0/0 and G0/0/1 are enabled.

10.2.3

Verify Interface Configuration

There are several commands that can be used to verify interface configuration. The most useful of these is the **show ip interface brief** and **show ipv6 interface brief** commands, as shown in the example.

```
R1#  show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.10.1    YES manual up                    up
GigabitEthernet0/0/1   209.165.200.225 YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#  show ipv6 interface brief
GigabitEthernet0/0/0       [up/up]
    FE80::201:C9FF:FE89:4501
    2001:DB8:ACAD:10::1
GigabitEthernet0/0/1       [up/up]
    FE80::201:C9FF:FE89:4502
    2001:DB8:FEED:224::1
Vlan1                      [administratively down/down]
    unassigned
R1#
```

10.2.4

Configuration Verification Commands

The table summarizes the more popular **show** commands used to verify interface configuration.

Commands	Description
`show ip interface brief` `show ipv6 interface brief`	The output displays all interfaces, their IP addresses, and their current status. The configured and connected interfaces should display a Status of "up" and Protocol of "up". Anything else would indicate a problem with either the configuration or the cabling.
`show ip route` `show ipv6 route`	Displays the contents of the IP routing tables stored in RAM.
`show interfaces`	Displays statistics for all interfaces on the device. However, this command will only display the IPv4 addressing information.
`show ip interface`	Displays the IPv4 statistics for all interfaces on a router.
`show ipv6 interface`	Displays the IPv6 statistics for all interfaces on a router.

Click each button to see the command output for each configuration verification command.

show ip interface brief | show ipv6 interface brief | show ip route | show ipv6 route | show interfaces | show ip interface | show ipv6 interface

```
R1# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.10.1    YES manual up                    up
GigabitEthernet0/0/1   209.165.200.225 YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

10.2.5

Syntax Checker - Configure Interfaces

Use this syntax checker to practice configuring the GigabitEthemet 0/0 interface on a router.

- Describe the link as 'Link to LAN'.
- Configure the IPv4 address as 192.168.10.1 with the subnet mask 255.255.255.0.
- Configure the IPv6 address as 2001:db8:acad:10::1 with the /64 prefix length.
- Activate the interface.

```
Enter global configuration mode.

R1#
```

Reset | Show Me | Show All